

Proyecto Integrador

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1 Desarrollo

Matriz de rigidez de una barra

$$\begin{bmatrix} q_{N_{x'}} \\ q_{N_{y'}} \\ q_{N_{z'}} \\ q_{F_{x'}} \\ q_{F_{y'}} \\ q_{F_{z'}} \end{bmatrix} = \begin{bmatrix} \frac{AE}{L} & 0 & 0 & -\frac{AE}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{AE}{L} & 0 & 0 & \frac{AE}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \begin{bmatrix} d_{N_{x'}} \\ d_{N_{y'}} \\ d_{N_{z'}} \\ d_{F_{x'}} \\ d_{F_{y'}} \\ d_{F_{z'}} \end{bmatrix}$$

$$\mathbf{q} = \mathbf{k}'\mathbf{d}$$

Matriz de desplazamientos globales y locales

$$\begin{bmatrix} d_{N_{x'}} \\ d_{N_{y'}} \\ d_{N_{z'}} \\ d_{F_{x'}} \\ d_{F_{y'}} \\ d_{F_{z'}} \end{bmatrix} = \begin{bmatrix} \lambda_x & \lambda_y & 0 & 0 & 0 & 0 \\ -\lambda_y & \lambda_x & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda_x & \lambda_y & 0 \\ 0 & 0 & 0 & -\lambda_y & \lambda_x & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} D_{N_x} \\ D_{N_y} \\ D_{N_z} \\ D_{F_x} \\ D_{F_y} \\ D_{F_z} \end{bmatrix}$$

$$\mathbf{d} = \mathbf{T}\mathbf{D}$$

Cosenos directores

$$\lambda_x = \cos \theta_x = \frac{x_F - x_N}{L} = \frac{x_F - x_N}{\sqrt{(x_F - x_N)^2 + (y_F - y_N)^2}}$$
$$\lambda_y = \cos \theta_y = \frac{y_F - y_N}{L} = \frac{y_F - y_N}{\sqrt{(x_F - x_N)^2 + (y_F - y_N)^2}}$$

Matriz de transformacion de la fuerza

$$\begin{bmatrix} Q_{N_x} \\ Q_{N_y} \\ Q_{N_z} \\ Q_{F_x} \\ Q_{F_y} \\ Q_{F_z} \end{bmatrix} = \begin{bmatrix} \lambda_x & -\lambda_y & 0 & 0 & 0 & 0 \\ \lambda_y & \lambda_x & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda_x & -\lambda_y & 0 \\ 0 & 0 & 0 & \lambda_y & \lambda_x & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} q_{N_{x'}} \\ q_{N_{y'}} \\ q_{N_{z'}} \\ q_{F_{x'}} \\ q_{F_{y'}} \\ q_{F_{z'}} \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{T}^T \mathbf{q}$$

Matriz de rigidez global

$$\begin{aligned}\mathbf{q} &= \mathbf{k}'\mathbf{T}\mathbf{D} \\ \mathbf{Q} &= \mathbf{T}^T\mathbf{k}'\mathbf{T}\mathbf{D} \\ \mathbf{Q} &= \mathbf{k}\mathbf{D} \\ \mathbf{k} &= \mathbf{T}^T\mathbf{k}'\mathbf{T}\end{aligned}$$

Para obtener los resultados se resuelve

$$\mathbf{Q} = \mathbf{k}\mathbf{D}$$

Para los resultados de cada barra luego se calcula

$$\mathbf{q} = \mathbf{k}'\mathbf{T}\mathbf{D}$$

Gracias

¡Gracias por tu atención!